

## **Concept Note**

### **Project Title**

**PathPilot AI: An Agentic Multi-Agent Career Intelligence System using LangGraph and Generative AI**

### **Selected SDG**

**SDG 4 – Quality Education**

### **Reason for Selecting this SDG**

Quality education extends beyond formal schooling and includes access to personalized guidance that helps individuals develop relevant skills and prepare for future careers. Many students and fresh graduates struggle to identify the skills required for their desired careers, choose appropriate learning resources, and build strong portfolios. Existing career guidance platforms generally provide static recommendations and fail to adapt to individual goals and skill levels.

This project supports SDG 4 by providing an AI-powered personalized career guidance platform that helps learners make informed decisions and continuously improve their employability.

### **Problem Statement**

Students often lack personalized career guidance while preparing for technical roles. They are uncertain about the skills they need, which learning resources to follow, what projects to build, and how to prepare for interviews. This results in inefficient learning, wasted effort, and poor career planning.

Current career recommendation systems generally offer generic advice and do not intelligently coordinate multiple tasks required for complete career planning.

### **Proposed Solution**

**PathPilot AI** is an **Agentic Multi-Agent Career Intelligence System** developed using **Google Gemini**, **LangGraph**, and **Streamlit**.

Unlike traditional chatbots, PathPilot AI employs a **Supervisor Agent** that intelligently plans and routes tasks among specialized AI agents. Each agent performs a dedicated responsibility

such as skill-gap analysis, learning resource recommendation, project suggestion, or interview preparation. LangGraph manages the workflow through shared state management and conditional routing, enabling dynamic collaboration between agents.

The system generates a comprehensive personalized learning roadmap based on the user's career goal, existing skills, and preferred learning timeline.

### **Objectives**

- Provide personalized career guidance using Generative AI.
- Identify skill gaps based on career goals.
- Recommend curated learning resources.
- Suggest industry-relevant portfolio projects.
- Generate interview preparation questions.
- Demonstrate Agentic AI using LangGraph-based multi-agent orchestration.

### **Expected Outcome**

The proposed system enables students to make informed learning decisions, reduce confusion in career planning, and develop structured learning paths aligned with industry requirements. By integrating Generative AI with an Agentic AI workflow, the solution offers a scalable and intelligent approach to personalized education, directly contributing to the objectives of **SDG 4: Quality Education**.